IoT-Based Patient Monitoring System - Connections & Schematic

1. ESP32 Pinout & Connections

Component	ESP32 Pin	Other Connection Notes
MAX30100	SDA -> GPIO21, SCL -> GPIO22	VCC -> 3.3V or 5V, GND -> GND
DHT22	Data -> GPIO19	VCC -> 3.3V or 5V, GND -> GND, add 10k pull-up resistor
DS18B20	Data -> GPIO18	VCC -> 3.3V or 5V, GND -> GND, 4.7k pull-up resistor
OLED 0.96"	SDA -> GPIO21, SCL -> GPIO22	VCC -> 3.3V or 5V, GND -> GND
Warning LED	GPIO2 -> Anode (+)	Cathode (-) -> GND, 220 Ohm resistor in series
Buzzer	GPIO15 -> +	- -> GND

2. Wiring Notes

1. I2C Bus Sharing
   - MAX30100 and OLED both use SDA -> GPIO21 and SCL -> GPIO22. I2C allows multiple devices on the same bus.

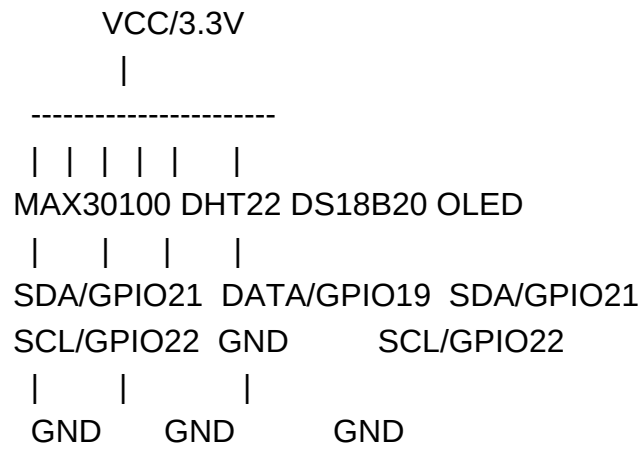
2. Pull-up Resistors
   - MAX30100 breakout usually has internal pull-ups; otherwise, 4.7k pull-up on SDA and SCL.
   - DS18B20 requires 4.7k pull-up between DATA and VCC.
   - DHT22 requires 10k pull-up between DATA and VCC.

3. LED & Buzzer
   - LED uses a 220 Ohm resistor.
   - Buzzer can be active; connect directly to GPIO15 and GND.

3. Schematic Layout Idea

- Left side: Sensors (MAX30100, DHT22, DS18B20)
- Right side: OLED display
- Bottom: Warning LED + Buzzer
- Power: All devices share 3.3V (or 5V) and GND

ASCII Diagram:



Warning LED -> GPIO2 -> 220 Ohm -> GND
Buzzer -> GPIO15 -> GND

Notes:
- Ensure proper pull-up resistors for DHT22 and DS18B20.
- LED and buzzer are optional but provide visual and audio fever alerts.
- All sensors and display share common GND and power rails.